1- dars (Model Improvement)

1. Xulosa va yangi o'yimiz talablari
2. Model Improvement uchun maqsad
3. Metrics -> Advanced Metrics
4. Threshold tuning
5. Probability of class $< 1 \Rightarrow$ y-pred[0] = 1 shuni 0 foizda extimalini
6. Threshold tuning and Advanced Metrics topish.
7. Vazifa.

Modular programming o'yligida biz deployment qilamiz. Hozirgi ishlarimiz, hozirgi loyihalarimiz, hozirgacha model building qilgan ishlarimiz nimada? Hozirgi loyihalarimiz faylga teng holos. Hozir biz fayl yaratayapmiz. Demak biz hozirgi loyihalarimizni bitta ipynb fayl desak -> bu product maxsulot emass!!!

Haqiqiy maxsulot bo'lishi u-n, masalan biz bir telefon do'koniga kirdik va u yerda telefon maketi bo'ladi lekin u telefon emas. Biz shu maketni yasadik endi esa o'sha maketdan -> telefonga o'tib olishimiz k-k. Ya'ni xaqiqiy maxsulotga boshqalar ham ishlata olishi mumkin bo'lgan. Deployment o'yligi bo'ladi.

Model Improvement o'yi uchun maqsad!

CamScanner

No.

tufayli stabilligi ölib qolmaslik k.-k.
Model mukammal bölishi k-k. Mukammal
deganda qanaqa → tögri xato kamchilik/dan
holi bölmaydi, turli hil xato va kamchi-
liklar bölishi tabiiy, model agar featurelarimiz
malum bir muddat ötgandan keyin qadri ka-
mayishi mumkin. Yoki umuman oldin Datasetimiz
körmagan datalar bly prediction qilayotgan
juda manosiz javob ~~chiqay~~ chiqarib yuboraydi.
Shu = u-n ham Robustness & Cross-Validation Strategy-
ni körib chiqishimiz k-k.

- Problem Solving (6-kyegacha)
- 2ta Project: Albatta rivojlangan model = projecti
bölishi k-k.

Metrics → Advanced Metrics
Hozircha Regressionni unutib turaylik faqat
Classification haqida gaplashamiz. Yana ham
Classification Report haqida gaplashamiz.

Classificationda: Confusion Matrix bor edi.

- TP === Model positive classlarni necha marta
tögri predict qilgan (Qerol aniqlandi)
Aslida 1 edi natija ham 1 chiqdi.
- FP === Model positive predict qilgan lekin
negative bölgan (Öyinchoq qerol xaqiqiy qerol deb
aniqlandi) → Aslida 0 edi lekin model buni 1
deb predict qildi.

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- TN === Model negative classni tõgri predict gilgan (Kiyimlar sawsiz deb topildi) Aslida 0 edi 0 deb topdi.
- FN === Model negative predict gilgan lekin positive bõlgan (Qarol tekshiruvdan õtib ketib goldi)

Demak, etibor bergan bõlsanolar, classification-reportni topayotgarimizda eng asósiy basgich bu confusion matrix edi. Yani õsha loyihadagi õsha topayotgan, chiqarayotgan soni.

Classification va Regressionlar = metrics/ini õrtasidagi farqi Nimada?

I. Regressionda → bu input va output qiymatlarini biz-biriga ganchalik yaqinligi bilan ölchanadigan bõlsa.

II. Classificationda esa → bu õsha tõgri va notõgri predict gilinganlarni soni bõyicha hisoblanadi. Qancha kõp TP chiqsa umumiy classification-report yuqori natija berar edi. Qancha False/kõpayib ketsa past natija berar edi.

Accuracy score → Model tõgri topgan ~~sonlar~~ prediction/ulushi

- Accuracy score:
  I. Model tõgri topgan prediction/ulushi
  II. Accuracy score = (TP+TN)/TOTAL
  III. Agar 98/100 toi tõgri misol ishlandi, 2ta ishlanmadi. Lekin shu 2ta imtiyonda tushib goldi

No.

- Yani accuracy_score qayerda xatolik bor ekanligini bersitadi!

Precision

- Ul model positivllarni ichida qanchasini tõõri predict qilgan.
↳ Masalan 70 ta positive bor 100 tani ichida shu 70tadan nechtasini tõõri predict qildi.
Presision formulasi = TP / TP + ~~FN~~ FP

- Agar odam xaqiqatdan imtiscondan õtishga loyiq bõlgandagina pass aks xolda faill /// Yoki faqatgina aybdorligi 100% aniq bõlgan odamgina jazolanishi.

- Important / Dangerous Case: FP xawfli bõlganda (aybsiz odamni aybdor qilish, soglom odamni kasal deyish) → Shunday loyihalar borki xato qilish juda qimatga tushishi mumkin shunda biz → presisionni oshirishimiz va unga e'tibor berishimiz shart !!!

- Focus: FP ni kõpayib ketishini oldini olishimiz kerak (kamroq FP)

- Xulosa: Agar FP xawfli (false alarms) bõladigan xolatlarda Precissionni yuqori bõlishiga xarakat qilishimiz k-k.
High Threshold (0.8)

Recall: ML model qancha TP ni topganligini o'lchaydi

- Positivelar qolib ketmasligi kerak.

Formulasi: TP / TP+FN

- Focus: FN lar ko'payib ketishini oldini olishimiz kerak → FN ko'payib ketsa Recall ↘ ketadi.

! Demak → FN larni kamaytirishimiz kerak!

Masalan:

Actual	Predicted
0	1
1	0

asli 1 bunday predicted qilsa ya 0 lar oshib ketsa 1 larni soni kamayib ketib qoladida. Chunki 100 ta qiymatdan 1 tasi shunday xato ketdimi sizlarda 99 ta qoladida. Lekin hudde shunday xatolik bolgani bilan, TP ni ham kamaytirib yuboradi. va pirovardida Recall ↘ ham kamayib ketadi.

low Threshold (0.3)

F1 score

- Harmonic mean of $\boxed{\text{Precision}}$ and $\boxed{\text{Recall}}$

Formulasi: $F1\ score = \frac{2 * Precision * Recall}{Precision + Recall}$

- Balance ni taminlaydi

Xulosa:

- Model 98% accuracy bersa xam foydasiz model bolishi mumkin.
- Yani kuchli 98% lik model xam notogri qaror chiqarishi mumkin. Shunda faqat accuracy qq. suyanib qolmasdan precision va recall b/n ham ish olib borishimiz kerak

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Threshold tuning

- Threshold tunning == decision controller
↳ Yani bitta tushuncha bor: Yaxshi model yaratish bor va nazorat qiliradigan model yaratish bor. Yani modellarimiz yaxshi bolishi mumkin lekin biz nazorat qila olishligimiz kerak boladi.

- Threshold == shifokorning bemorni kasal yoki kasal emasligini belgilash chegarasi.
→ Masalan isitma 38°C dan oshsagina dori ber deya qqar threshold qoymasa temperatura yana ham tushib ketishi m-n.

- Uni ozgartirish sayotini ozgartiradi.
Demak Thresholdni toori qoya olish demak modelimizni yaxshi ishlatishni ham ta'minlaydi ekan.

⟹ Quyidagi asosiy qoidalarga koora threshold tanlanadi:
- Business goal: Biznes maqsad
- Risk tolerance: Tavakkalchilikka chidamlilik.
- Cost of errors: (FP vs FN)
- Class imbalance:

Threshold tuning and Advanced Metrics

- Classification
- Prediction
- 0 → Negative
- 1 → Positive

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▲ probs = model.predict_proba(X_test)

print("Raw probabilities:", probs[10])

Negative ↩ ↪ Positive

positive_prob = probs[0][1]

print(f"Model confidence for positive class: {positive_prob*100:.2f}%")

Ya'ni chiqayotgan javobni ishonchlilik foizi. miqdori.
Precision, accuracy lar bu model = ishlashi aniqligi.
Javobni chiqarish aniqligi emas. predict_proba()
-> bu javobni chiqarish aniqligi.
Model yaxshi ishlashi u-n., ozi. tog'ri aniqlikda ishlashi k-k. Va mantiqli javob berishi k-k.

- Threshold: (probability ≥ threshold)
- Qachon javobni qabul qilamiz.
- Javobni to'g'ri deb qabul qiladigan chiziq
- Low (0.3) → 0,2 farq → Recall ↗
- Medium (0.5)
- High (0.8) — 0.3 farq → Precision ↗

▲
```
import numpy as np

probs = np.array([
    [0.10, 0.90],
    [0.30, 0.70],
    [0.40, 0.60],
    [0.70, 0.30],
    [0.80, 0.10],
])
```

```
for i, p in enumerate(probs):
    print(f"Sample {i}: Raw probabilities:", p)
    positive_prob = p[1]
    print(f"Model confidence for positive class: {positive_prob*100:.2f}%")
```

No.

Thresholddan foydalanish va tadbiq qilish.

```
import numpy as np
from sklearn.metrics import precision_score, recall_score

# Example model probabilities
probs = np.array([0.90, 0.70, 0.60, 0.30, 0.20])

# true labels
y_true = np.array([1, 1, 0, 0, 1])
print("Predicted Probabilities:", probs)
print("True Labels:", y_true)

# ========================
# 1. LOW THRESHOLD (0.3)
# ========================
low_thr = 0.3
y_pred_low = (probs >= low_thr).astype(int)

print("\n === LOW THRESHOLD (0.3) ===")
print("Predictions:", y_pred_low)
print("Precision:", precision_score(y_true, y_pred_low)
print("Recall:", recall_score(y_true, y_pred_low)

# ========================
# 2. MEDIUM THRESHOLD (0.5)
# ========================
med_thr = 0.5
y_pred_med = (probs >= med_thr).
# HUDDI SHUNI YOZAMAN FAQAT "\n == 2
MEDIUM THRESHOLD (0.5) == 2"
```

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```
# = = = = = = = = = = = = =
3. HIGH THRESHOLD (0.8)
# = = = = = = = = = = = = = = = =
high_thr = 0.8
y_pred_high = (probs >= high_thr).astype(int)

# YUQORIDAGI YANA YOZILADI
```